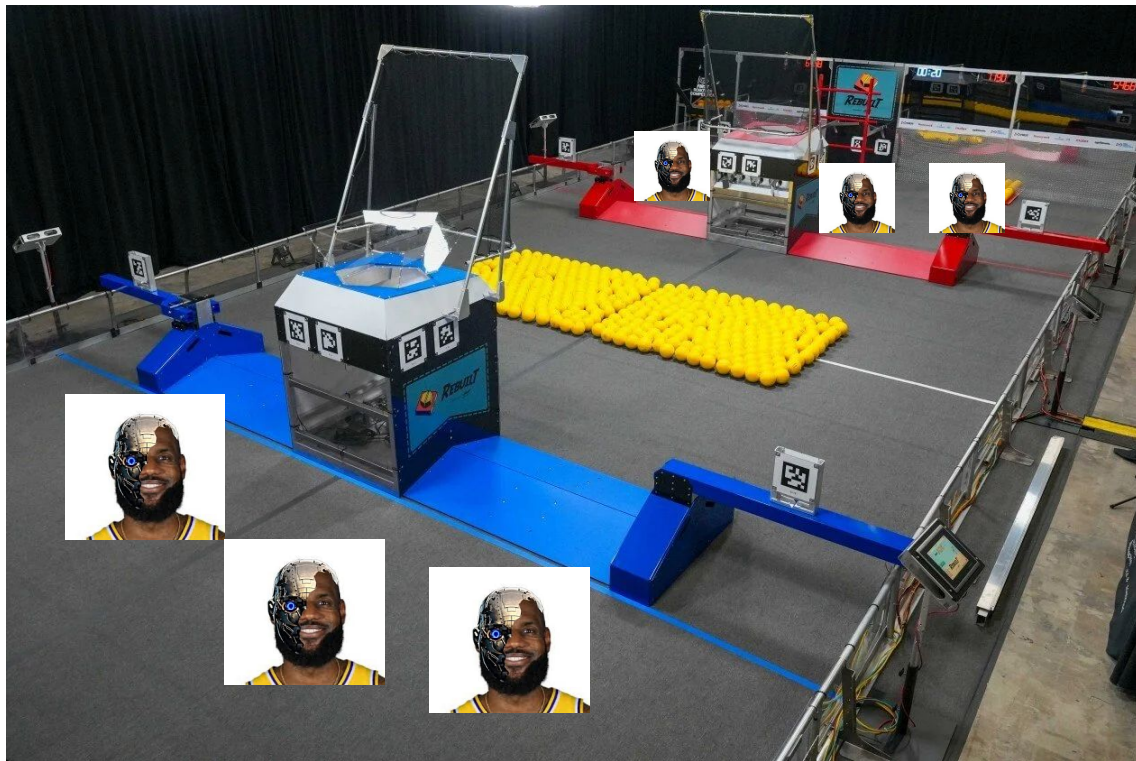




Click to add subtitle

Chinese remainder theorem

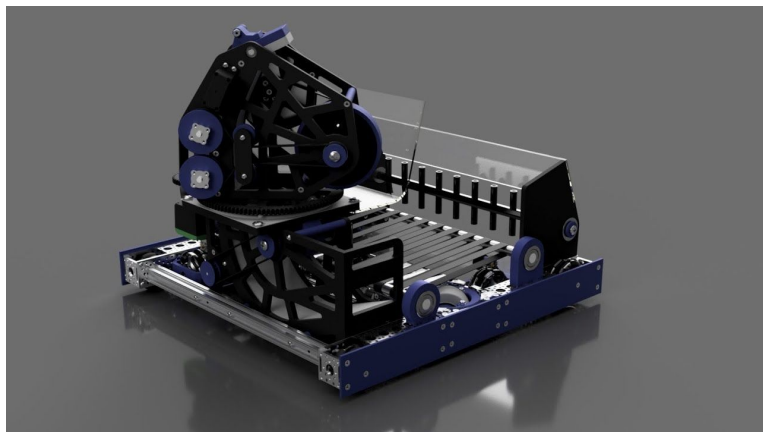
FRC robotics 2026



FRC hardware



Turret



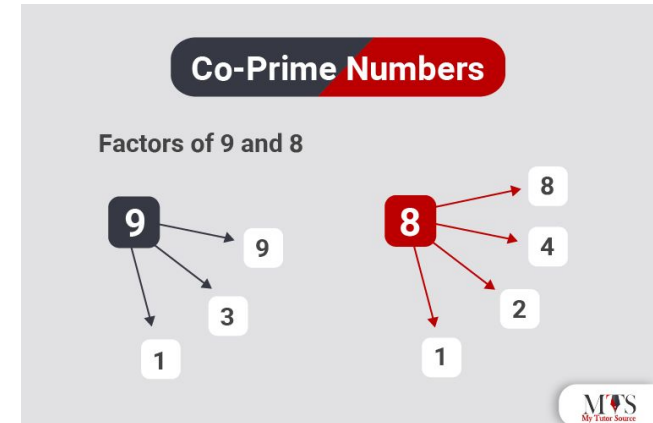
How to always know which way turret pointed, Allways?



Chinese remainder theorem (or Sunzi's theorem)

If you have a set of pairwise coprime integers (numbers that share no common factors other than 1), the CRT states that a system of simultaneous congruences has a unique solution.

If you know that a value x leaves certain remainders when divided by different numbers, you can figure out exactly what x is, provided x is within a certain range.



example

5 and 7 and 35

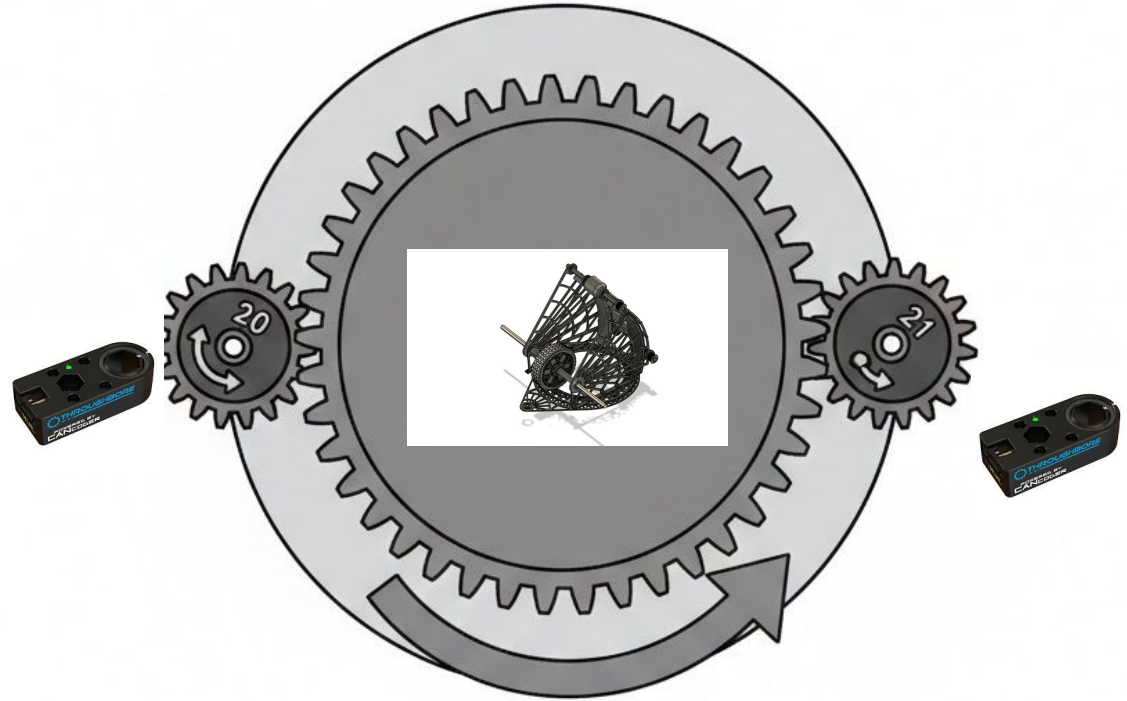
1	2	3	4	0	1	2	3	4	0	1	2	3	4	0	1	2	3	4	0	1	2	3	4	0	1	2	3	4	0	1	2	3	4	0
1	2	3	4	5	6	0	1	2	3	4	5	6	0	1	2	3	4	5	6	0	1	2	3	4	5	6	0	1	2	3	4	5	6	0



Application irl

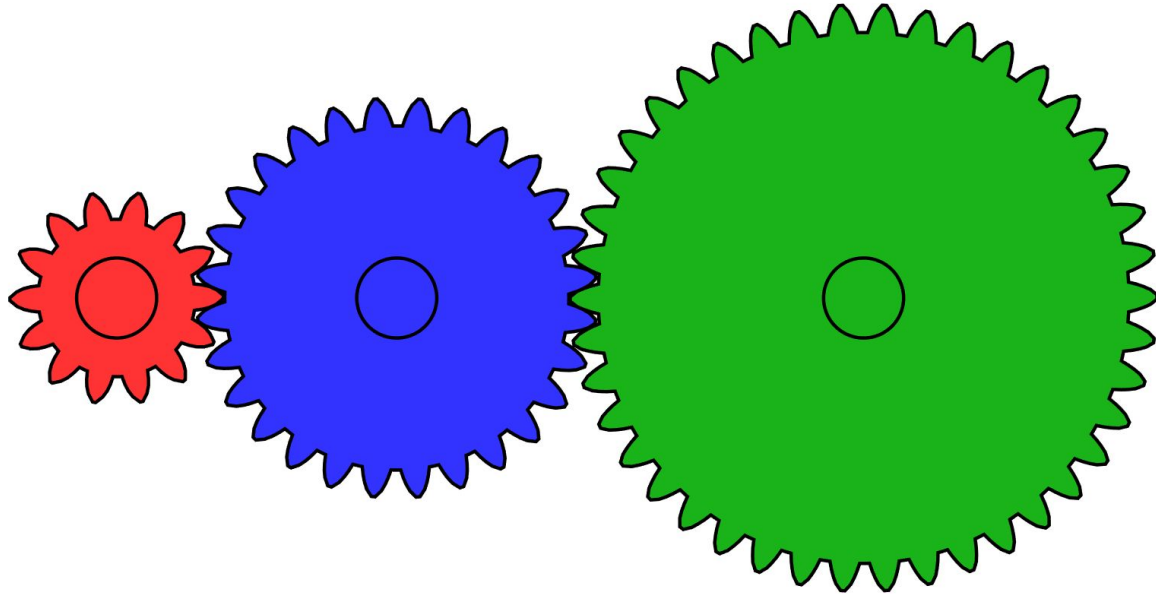
Wow!

Glory to the CCP!



Good news! most gears are manufactured coprime!

ensures that the same teeth doesn't end up in a cycle of which ridges it goes through, reduces wear and noise.



Bad news, most gears are manufactured to be coprime

Turret gear must be manufactured

